

Mark schemes

Q1.

$$y_{n+1} \approx y_n + h f(x_n)$$

OE

$$h y'(1) = 0.1 \times y'(1) (=0.05)$$

Attempt to find $h y'(1)$. PI by eg 3.05 for $y(1.1)$

M1

$$y(1.1) \approx 3 + 0.05 = 3.05$$

A1

$$y(1.2) \approx y(1.1) + 0.1 \times y'(1.1) = 3.05 + 0.1 \times y'(1.1)$$

$$\approx 3.05 + 0.1 \times \frac{1.1}{1+1.1^3} \left(= 3.05 + 0.1 \times \frac{1100}{2331} \right)$$

Attempt to find $y(1 + 0.1) + 0.1 \times y'(1 + 0.1)$ must see evidence of calculation if correct ft $[0.047.. + c's y(1.1)]$ value not obtained

m1

$$\approx 3.05 + 0.047(19.....)$$

OE; ft on $[0.047.. + c's y(1.1)]$ value; PI

A1F

$$\approx 3.0972 \text{ (to 4 d.p.)}$$

Must be 4 dp.

©2025 Exam Papers Practice. All rights reserved.

A1

[5]

Q2.

$$(a) \quad y(3.2) = y(3) + 0.2\sqrt{2 \times 3 + 5}$$

M1

$$= 5 + 0.2 \times \sqrt{11}$$

A1

$$= 5.66332... = 5.6633 \text{ to 4dp}$$

Condone > 4dp

A1

(b) $y(3.4) = y(3) + 2(0.2) \{f[3.2, y(3.2)]\}$

M1

$$\dots = 5 + 2(0.2)\sqrt{2 \times 3.2 + 5.6633\dots}$$

Ft on cand's answer to (a)

A1F

$$= 5 + (0.4)\sqrt{12.0633\dots}$$

$$= 6.389 \text{ to 3dp}$$

CAO Must be 6.389

A1

3

[6]

Q3.

(a) $y(1.1) = y(1) + 0.1 \left[\frac{2-1}{4+1} \right]$

M1A1

$$= 2 + 0.02 = 2.02$$

A1

3

(b) $y(1.2) = y(1) + 2(0.1)\{f[1.1, y(1.1)]\}$

M1

$$= 2 + 2(0.1) \left[\frac{2.02-1.1}{2.02^2+1.1} \right]$$

ft on c's answer to (a)

A1F

$$= 2.035518\dots = 2.036 \text{ to 3dp}$$

CAO Must be 2.036

A1

3

[6]

Q4.

Attempt at $0.5 \times y'(2) (= 0.25)$

Other variations are allowed

M1

$$y(2.5) \approx 3.25$$

A1

$$y(3) \approx 3.25 + 0.5 y'(2.5)$$

m1

$$\approx 3.25 + 0.2357(0)$$

PI; OE; ft c's value for y(2.5)

A1F

$$\approx 3.4857$$

4 dp needed

A1

[5]

Q5.

(a) $y_1 = 2 + 0.1 \times [3 \ln(2 \times 3 + 2)] = 2 + 0.3 \ln 8$
 $= 2.6238(3\dots)$

M1A1

$$y(3.1) = 2.6238 \text{ (to 4 dp)}$$

Condone greater accuracy

A1

3

(b) $k_1 = 0.1 \times 3 \ln 8 = 0.6238(32\dots)$

PI ft from (a), 4 dp or better

B1F

$$k_2 = 0.1 \times f(3.1, 2.6238(32\dots))$$

M1

$$\dots = 0.1 \times 3.1 \times \ln 8.8238(32\dots)$$

$$[= 0.6750(1\dots)]$$

PI; ft on $0.1 \times 3.1 \times \ln[6.2 + \text{answer}(a)]$

A1F

$$y(3.1) = 2 + \frac{1}{2} [0.6238(3\dots) + 0.6750(1\dots)]$$

m1

$$= 2.6494(2\dots) = 2.6494 \text{ to 4 dp}$$

CAO Must be 2.6494

A1

5

Q6.

First increment is $0.1 \times 2 (= 0.2)$
variations possible here

M1

So next value of y is 3.2
PI

A1

Second inc't is $0.1(1 + 1.1^3) = 0.2331$
PI

m1A1

Third inc't is $0.1(1 + 1.2^3) = 0.2728$
PI

A1

So $y \approx 3.7059 \approx 3.706$
ft one numerical error

A1F

[6]

Q7.

(a) $y(1.1) = y(1) + 0.1[1 + 3 + \sin 1]$

M1A1

$= 1 + 0.1 \times 4.84147. = 1.4841(47..) = 1.4841 \text{ to 4 dp}$
Condone > 4 dp

A1

3

(b) $y(1.2) = y(1) + 2(0.1)\{f[1.1, y(1.1)]\}$

M1

$.... = 1 + 2(0.1)\{1.1 + 3 + \sin[1.4841(47..)]\}$
Ft on cand's answer to (a)

A1F

$= 2.019 \text{ to 3 dp}$

CAO Must be 2.019

Note: If using degrees max mark is 4/6 ie M1A1A0; M1A1FA0

Q8.

First increment is 0.2, so $y \approx 1.2$

PI; variations possible here

B1B1

Second increment is $0.2\sqrt{1+0.2^2}$

M1

... $\approx 0.203\ 961$, so $y \approx 1.403\ 96$

A1 if accuracy lost; ft num error

A2,1F

[5]

Q9.

(a)
$$y_1 = 3 + 0.2 \times \left[\frac{1^2 + 3^2}{1 + 3} \right]$$

M1A1

$$= 3.5$$

A1

3

(b) $k_1 = 0.2 \times 2.5 = 0.5$

PI ft from (a)

B1Ft

$k_2 = 0.2 \times f(1.2, 3.5)$

ft on (a)

M1

$$\dots = 0.2 \times \frac{1.2^2 + 3.5^2}{1.2 + 3.5} = 0.5825(53 \dots)$$

PI condone 3 dp

A1Ft

$$y(1.2) = y(1) + \frac{1}{2} [0.5 + 0.5825(53 \dots)]$$

m1

$$= 3.54127 \dots = 3.5413 \text{ to 4dp}$$

ft one slip

If answer not to 4 dp withhold this mark

A1Ft

5

[8]

Q10.

(a) $y(3.1) = y(3) + 0.1 \sqrt{3^2 + 2 + 1}$

M1A1

$$= 2 + 0.1 \times \sqrt{12} = 2.3464 \text{ (10..)} \\ = 2.3464$$

Condone > 4 dp if correct

A1

3

(b) $y(3.2) = y(3) + 2(0.1)[f(3.1, y(3.1))]$

M1

$$\dots = 2 + 2(0.1) \left[\sqrt{(3.1)^2 + 2.3464 + 1} \right]$$

ft on candidate's answer to (a)

A1F

$$\dots = 2 + 0.2 \times 3.599499\dots = 2.719(89\dots) \\ = 2.720$$

CAO Must be 2.720

A1

3

[6]

Q11.

0.01 (2^1) added to value of y

Variations possible here

M1

So $y(1.01) \approx 4.02$

PI

A1

Second increment is $0.01(2^{1.01})$

m1

... $\approx 0.020\ 139$

A1

So $y(1.02) \approx 4.040\ 14$

A1

[5]

Q12.

(a) $y(2.1) = y(2) + 0.1[2^2 - 1^2]$

M1A1

$$= 1 + 0.1 \times 3 = 1.3$$

A1

3

(b) $y(2.2) = y(2) + 2(0.1)[f(2.1, y(2.1))]$

M1

$$\dots = 1 + 2(0.1)[2.1^2 - 1.3^2]$$

Ft on cand's answer to (a)

A1ft

$$\dots = 1 + 0.2 \times 2.72 = 1.544$$

CAO

A1

3

EXAM PAPERS PRACTICE [6]

©2025 Exam Papers Practice. All rights reserved.

Q13.

(a) $y(1.05) = 0.6 + 0.05 \times [\ln(1 + 1 + 0.6)]$

M1A1

$$= 0.6477\ (7557\dots) = 0.6478\ \text{to 4dp}$$

Condone > 4 dp

A1

3

(b) $k_1 = 0.05 \times \ln(1 + 1 + 0.6) = 0.0477\ (75\dots)$

PI

M1

ft candidate's evaluation in (a)

A1F

$$k_2 = 0.05 \times f(1.05, 0.6477\dots)$$

$$\dots = 0.05 \times \ln(1 + 1.05^2 + 0.6477\dots)$$

M1

$$\dots = 0.0505(85 \dots)$$

PI

A1F

$$y(1.05) = y(1) + \frac{1}{2}[k_1 + k_2]$$

Dep on previous two Ms and numerical values for k's

m1

$$= 0.6 + 0.5 \times 0.09836\dots$$

$$= 0.6492 \text{ to 4dp}$$

Must be 4 dp... ft one slip

A1F

6

[9]

Q14.

(a) $y_1 = 2 + 0.1 \times \sqrt{1^2 + 2^2 + 3}$

M1

$$y(1.1) = 2 + 0.1 \times \sqrt{8}$$

©2025 Exam Papers Practice. All rights reserved.

A1

$$y(1.1) = 2.28284\dots = 2.2828 \text{ to 4dp}$$

A1

3

(b) $k_1 = 0.1 \times \sqrt{8} = 0.2828$

PI

M1

A1ft

$$k_2 = 0.1 \times f(1.1, 2.2828\dots)$$

M1

$$= 0.1 \times \sqrt{9.42137\dots} = 0.3069(425\dots)$$

PI

A1

$$y(1.1) = y(1) + \frac{1}{2} [0.28284... + 0.30694...]$$

m1

$$2.29489... = 2.2949 \text{ to 4dp}$$

A1

6

[9]

Q15.

(a) (i) $y(1.1) = y(1) + 0.1[1\ln 1 + 1/1]$

M1A1

$$= 1 + 0.1 = 1.1$$

A1

3

(ii) $y(1.2) = y(1) + 2(0.1)[f(1.1, y(1.1))]$

M1A1

$$... = 1 + 2(0.1)[1.1 \ln 1.1 + (1.1/1.1)]$$

On answer to (a)(i)

A1 ✓

$$... = 1 + 0.2 \times 1.104841198 ...$$

$$... = 1.22096824 = 1.221 \text{ to 3 dp}$$

CAO

A1

4

(b) (i) IF is $e^{\int -\frac{1}{x} dx}$
Condone $e^{\int \frac{1}{x} dx}$ for M mark

M1

$$= e^{-\ln x}$$

A1

$$= e^{\ln x^{-1}} = x^{-1} = \frac{1}{x}$$

AG (be convinced)

(b)(i) Solutions using the printed answer must

A1

3

(ii) $\frac{d}{dx} \left(\frac{y}{x} \right) = \ln x$

M1A1

$$\frac{y}{x} = \int \ln x \, dx = x \ln x - \int x \left(\frac{1}{x} \right) dx$$

Integration by parts for $x^k \ln x$

M1

$$\frac{y}{x} = x \ln x - x + c$$

Condone missing c.

A1

$$y(1) = 1 \Rightarrow 1 = \ln 1 - 1 + c$$

Dependent on at least one of the two previous M marks

m1

$$\Rightarrow c = 2 \Rightarrow y = x^2 \ln x - x^2 + 2x$$

OE eg $\frac{y}{x} = x \ln x - x + 2$

A1

6

©2025 Exam Papers Practice. All rights reserved.

(iii) $y(1.2) = 1.222543... = 1.223 \text{ to 3dp}$

B1

1

[17]

Q16.

1st increment is $0.2 \lg 2 \dots$

or $0.2 \lg 2.1$ or $0.2 \lg 2.2$

M1

$$\dots \approx 0.06021$$

PI

A1

$$x = 2.2 \Rightarrow y \approx 3.06021$$

PI; ft numerical error

A1ft

2nd increment is 0.2 lg 2.2
consistent with first one

m1

... ≈ 0.06848

PI

A1

$x = 2.4 \Rightarrow y \approx 3.12869 \approx 3.129$
ft numerical error

A1ft

6

[6]

Q17.

(a)
$$y_1 = 2 + 0.1 \times \left[\frac{1^2 + 2^2}{1 \times 2} \right]$$

$$= 2 + 0.1 \times 2.5 = 2.25$$

M1A1

A1

3

(b) $k_1 = 0.1 \times 2.5 = 0.25$

©2025 Exam Papers Practice. All rights reserved.

M1

PI ft from (a)

A1ft

$k_2 = 0.1 \times f(1.1, 2.25)$

M1

... $= 0.1 \times 2.53434... = 0.2534(34...)$
PI

A1ft

$$y(1.1) = y(1) + \frac{1}{2} [0.25 + 0.253434...]$$

m1

= 2.2517 to 4dp

If answer not to 4dp withhold this mark

A1ft

[9]

Q18.

$$x = 0.5 \Rightarrow y' = \sin 1$$

Max 4/6 if degrees used

M1

First increment is $0.1 \sin 1$

m1A1

$$x = 0.6 \Rightarrow y \approx 1.084$$

PI by correct answer at end

A1

Second increment is $0.1 \sin 1.2$

m1

$$x = 0.7 \Rightarrow y \approx 1.177 \approx 1.18$$

Ft error in y (0.6)

A1F

6

[6]

Examiner reports

Q1.

This opening question, which tested numerical methods, was answered correctly by the majority of students. The most common error was in the evaluation of $\frac{dy}{dx}$ at $x = 1.1$. Students who showed the substitution in addition to its evaluation generally gained an advantage over those who just wrote down the incorrect value in a table. Most students gave their final answer to the required degree of accuracy.

Q2.

Numerical solutions of first order differential equations are a good source of marks for all students. Once again, this topic was the best answered on the paper. The most common error involved an incorrect evaluation of $f[3.2, y(3.2)]$. Students lost fewer marks if the correct substitution into the formula had been shown before the incorrect evaluation. A slightly higher proportion of students than last year failed to give their final answer to the required degree of accuracy.

Q3.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. Once again, this topic was the best answered on the paper. The most common error involved an incorrect evaluation of $f[1.1, y(1.1)]$. Almost all candidates gave their final answer to the required degree of accuracy.

Q4.

This opening question was very well answered in the majority of cases. Many candidates showed their working in table form, but others provided slightly more detail of the methods used, which worked to their advantage if they made numerical errors. By far the most common way to lose a mark here was to perform three iterations rather than two, the candidates failing to pair off their x and y values correctly.

Q5.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates and this was the best answered question on the paper. Very few candidates mixed up the x and y values in applying the given formulae. Almost all candidates gave their final answers to the required degree of accuracy.

Q6.

Most candidates scored high marks on this opening question, though there were some who seemed to have no idea what to do. A mark was often lost by carrying out an unwanted fourth iteration. A small number of candidates used the upper boundaries of the intervals rather than the lower boundaries. This was accepted for full marks, but credit was lost if the candidate switched from one method to the other in the course of working through the solution.

Q7.

This was the best answered question on the paper. Numerical solutions of first order

differential equations continue to be a good source of marks for all candidates. The most common loss of marks was due to calculators being set in degree mode. It is worth recording that slightly more candidates than in recent series have slipped back into not showing the necessary working. Without such working, wrong answers cannot be awarded credit.

Q8.

The great majority of candidates were able to make a good start to the paper by giving a correct numerical solution to the differential equation. As on past papers, some candidates used the value of the derivative at the upper end of the interval rather than the lower end. This was perfectly acceptable for full marks, though these candidates gave themselves slightly more calculation to carry out as they were not using the very simple value of the derivative at $x = 0$. Some candidates carried out three iterations instead of two; these candidates lost a mark as well as giving themselves extra work.

Q9.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. This was the best answered question on the paper and it was pleasing to see full working with clear substitutions into relevant formulae. The vast majority of candidates obtained the correct answers, and errors were generally limited to incorrect evaluations rather than any lack of understanding of the methods or notation involved.

Q10.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. Although it was the best answered question on the paper, more candidates than usual mixed up the x and y values in applying the given formulae, or incorrectly used $f(3, 2)$ instead of $f(3.1, y(3.1))$ in the given formula in part (b). There were very few candidates who lost the final accuracy mark for failing to give their answer correct to three decimal places.

Q11.

©2025 Exam Papers Practice. All rights reserved.

This question provided most candidates with a further five marks. The most common error was to carry out three iterations instead of only two, which would usually cause the loss of only one mark as long as the working was fully shown; though of course the candidate may well have lost valuable time carrying out the unwanted calculations.

Q12.

Numerical solutions of first order differential equations continues to be a good source of marks for all candidates. This was the best answered question on the paper. Although almost all candidates obtained the correct answer to part (a), some less able candidates showed a lack of understanding of the notation used in the given formula in part (b).

Q13.

This question proved to be a good source of marks to all those candidates who showed sufficient working. There was some evidence of misreads, the most common one being “In $(1 + x^2 + y^2)$ ” for $f(x, y)$; for those candidates who explicitly showed this misread, the mark loss was minimal.

Q14.

This question which tested numerical solutions of differential equations was answered well with a large proportion of candidates gaining full marks. The most common error was the use of 2.1 instead of 2.2828... in finding k_2 . Although there were again candidates who failed to show working (by writing down the correct formula and then substituting in the correct values), the proportion doing so was significantly less than last year. The vast majority of candidates gave their final answers to the required degree of accuracy but more should be encouraged to show intermediate answers to more than the stated degree of accuracy.

Q15.

This question was a good source of marks for many candidates. The methods in part (a) were well understood and most candidates worked accurately to obtain the correct approximations. In part (b)(i) a high proportion of the candidates obtained the printed answer but some failed to show sufficient stages in their working. In part (b)(ii), weaker candidates could not integrate $\ln x$; such candidates frequently gave the answer as $1/x$. Those who inserted the constant of integration were generally able to apply the given boundary condition. Surprisingly, not all candidates who obtained the correct solution for part (b)(ii) were able to give the correct value for y in part (b)(iii).

Q16.

For most candidates this was an 'all-or-nothing' question. Either they knew what to do and usually carried it out efficiently, or they appeared to have no knowledge of this topic. Some worked to only three decimal places throughout the question and lost accuracy in the final answer, which cost them one mark. Others carried out three incrementations instead of only two, and again the penalty was only one mark for this misunderstanding.

Q17.

This question which tested numerical solutions of differential equations was the best answered question on the paper. The most common error was the use of 2.1 instead of 2.25 in finding k_2 . A significant number of candidates presented their values in the form of a well-labelled table but some included incorrect values with no indication where the error had occurred. As a result the method marks could not be awarded. Very few candidates failed to give their final answer to the required degree of accuracy.

Q18.

This question was reasonably well answered in general, but there were some common mistakes. One was to work in degrees throughout, despite the fact that the word 'radians' was printed in bold type. Another very common error was to continue the iterative process to the third iteration instead of stopping after the second. Some candidates failed to show enough working to make their method clear, while others attempted to solve the differential equation by integration rather than by the method required in the question.